

Management of Penetrating Neck Injuries: A New Paradigm for Civilian Trauma

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Purpose: Improvements in imaging technology, particularly computed tomographic angiography (CTA), have altered the management of patients with penetrating neck injuries. Although some centers still advocate routine exploration for all zone 2 neck injuries penetrating the platysma, many civilian centers in the United States have adopted a policy of selective exploration based on clinical and radiographic examination. The purpose of this retrospective study is to evaluate our 5-year experience with the management of penetrating neck injuries, to further elucidate the role of CTA in clinical decision-making, and to assess treatment outcome.

Patients and Methods: One hundred thirty-four consecutive patients were identified from the Legacy Emanuel Trauma Registry as having sustained penetrating neck injuries from 2000 to 2005. Using data collected from the Trauma Registry, as well as individual chart notes and electronic records, variables were collected and evaluated including age, gender, mechanism of injury, number of associated injuries, and the Injury Severity Score, Glasgow Coma Scale on admission, initial hematocrit, airway management techniques, diagnostic and therapeutic modalities, missed injuries, length of hospital stay, disposition, and outcome. Descriptive statistics were used to describe demographics, treatment, and outcome.

Results: One hundred twenty patients met the inclusion criteria, 55 of which had only superficial injuries that did not penetrate the platysma. The primary study group consisted of 65 patients who sustained more significant injuries that violated the platysma including deep, complex, and/or avulsive wounds, vascular injuries, injuries to the aerodigestive tract, musculoskeletal system, cranial nerves, or thyroid gland. The overall mortality rate for the 65 patients with injuries penetrating the platysma was 3.0% (n = 2). Complications occurred in 7 of the surviving 63 patients (10.7%): 2 patients with zone 3 internal carotid artery injuries developed hemispheric ischemic infarcts and hemiplegia; as well as other complications including: infection (n = 2); deep venous thrombosis (n = 1); aspiration pneumonia (n = 1); and hematoma (n = 1). All surviving patients except the 2 stroke patients eventually healed uneventfully without significant functional deficit. The use of CTA as a guide to clinical decision-making led to a significant decrease in the number of neck explorations performed and a virtual elimination of negative neck explorations.

Conclusion: The management of stable patients with neck injuries that penetrate the platysma has evolved at our institution into selective surgical intervention based on clinical examination and CTA and has resulted in minimal morbidity and mortality.

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J Oral Maxillofac Surg 65:691-705, 2007

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0278-2391/07/6504-0017\$32.00/0

doi:10.1016/j.joms.2006.04.044